# Fake and Real News Detection Using Machine Learning :An End-to-End Text Classification study

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## Abstract

**Fake news has become a significant challenge in the digital era because misleading information can spread opinion. This project develops an end-to-end machine learning system for automatically classifying news articles as Fake News or Real News. The project uses separate fake and real news datasets and follows a complete workflow including data loading, exploratory data analysis, duplicate and empty-text handling, text preprocessing, TF-IDF feature extraction, model development, validation, hyperparameter tuning, evaluation, comparison, model persistence, and prediction on unseen news articles. Logistic Regression, Multinomial Naive Bayes, and Linear Support Vector Machine models are trained and compared. The Linear SVM is further optimized using GridSearchCV with five-fold cross-validation. The tuned Linear SVM achieved the strongest overall performance and was selected as the final model, achieving 99.72% test accuracy and 99.71% F1-score. The trained model and vectorizer were saved and reused to classify unseen news text.**

## Keywords

Fake news detection, machine learning, text classification, TF-IDF, Linear SVM

## 1. Introduction

The rapid growth of digital media and online news platforms has made information easily accessible worldwide. However, it has also increased the spread of misleading and fabricated information. Fake news can influence public opinion, create confusion, and affect social and political decision-making. Manual verification of every article is difficult because of the volume of content generated daily. Machine learning offers an efficient solution by learning patterns from labelled examples and applying them to new articles. This project implements an end-to-end workflow for Fake and Real News Detection, including dataset preparation, preprocessing, TF-IDF transformation, multiple model development, comparison, tuning, evaluation, and unseen-data prediction.

## 2. Problem Statement

The problem addressed is the automatic classification of news articles into Fake News and Real News. Since news articles are unstructured textual data, they must be cleaned and transformed into numerical features before machine learning algorithms can process them. The objective is to develop a reliable classifier with strong accuracy, precision, recall, and F1-score.

## 3. Research Objectives

- Understand the Fake and Real News datasets.
- Combine datasets and create a binary classification problem.
- Perform data cleaning and text preprocessing.
- Perform exploratory data analysis.

- Transform text using TF-IDF.
- Train and compare multiple machine learning models.
- Evaluate using accuracy, precision, recall, and F1-score.
- Tune and validate the strongest model.
- Select a final model and predict unseen news.

## 4. Research Questions

- Can machine learning accurately classify news articles as fake or real?
- How effective is TF-IDF for fake news classification?
- Which evaluated model provides the strongest performance?
- Does hyperparameter tuning improve the selected model?
- Can the final model classify previously unseen news text?

## 5. Literature Review / Related Work

Text classification commonly uses statistical representations such as bag-of-words and TF-IDF together with algorithms including Naive Bayes, Logistic Regression, and Support Vector Machines. TF-IDF reduces the influence of very common words while preserving discriminative terms. Linear classifiers are particularly suitable for high-dimensional sparse text representations. This project applies these established approaches in a complete comparative workflow.

## 6. Research Gap

Many introductory fake-news projects train only one model and report only accuracy. This provides limited evidence about comparative model behavior. The present project addresses this gap by including data quality checks, text preprocessing, EDA, TF-IDF features, multiple-model comparison, five-fold validation, GridSearchCV tuning, final evaluation, model persistence, and unseen-news prediction.

## 7. Dataset Description

The Fake News dataset contains 23,481 records and the Real News dataset contains 21,417 records, giving 44,898 original combined records. The principal columns are title, text, subject, and date. Labels were assigned as Fake News = 0 and Real News = 1. After cleaning, 43,974 records remained.

| Class | Records |
|---|---|
| Fake News | 22,764 |
| Real News | 21,210 |
| Total | 43,974 |

## 8. Methodology

Separate Fake and Real News datasets were loaded and labelled before being combined. Duplicate and unsuitable records were handled and the text was cleaned. The cleaned dataset was split into training and testing subsets using a stratified 80:20 split. TF-IDF features were created from training text and applied consistently to testing text. Multiple classifiers were trained under the same workflow. The strongest baseline was tuned using GridSearchCV with five-fold cross-validation, after which the final model and vectorizer were saved for reuse.

## 9. System / ML Workflow

- Load Fake and Real News datasets
- Assign labels and combine datasets
- Inspect dataset and clean text
- Remove duplicates and unsuitable records
- Perform EDA
- Split data into training and testing sets
- Apply TF-IDF

- Train Logistic Regression, Multinomial Naive Bayes, and Linear SVM
- Evaluate and compare models
- Tune Linear SVM using GridSearchCV
- Select final model
- Save model and vectorizer
- Predict unseen news

## 10. Data Preprocessing

Preprocessing included combining the datasets, assigning numerical labels, checking text quality, removing duplicate or unsuitable records, cleaning and normalizing text, and preparing the final text representation for TF-IDF. The same preprocessing procedure was reused for unseen-news prediction to maintain consistency between training and deployment.

### 11. Exploratory Data Analysis (EDA)

EDA included inspection of dataset shape and columns, class distribution, missing values, duplicate records, and representative news content. After cleaning, the dataset contained 22,764 Fake News records and 21,210 Real News records, indicating a relatively balanced binary classification dataset.

### 12. Feature Engineering / Feature Selection

The main representation was TF-IDF with a maximum of 50,000 features and an n-gram range of (1,2). Unigrams capture individual words while bigrams capture short phrases, enabling the models to learn more informative textual patterns. No separate feature-selection algorithm was applied because the TF-IDF vocabulary was already limited to the selected feature space.

### 13. Model Development

Three models were developed: Logistic Regression as a strong linear baseline, Multinomial Naive Bayes as a standard probabilistic text classifier, and Linear SVM for high-dimensional sparse text data. The Linear SVM produced the strongest baseline performance and was selected for tuning.

| Model | Reason for Inclusion |
|---|---|
| Multinomial Naive Bayes | Common baseline for non-negative text features and text classification. |
| Logistic Regression | Strong linear baseline for binary classification and sparse features. |
| Linear SVM | Suitable for high-dimensional sparse text data. |

### 14. Experimental Setup

The project was implemented using Python with Pandas, NumPy, Matplotlib, Seaborn, scikit-learn, and Joblib. The cleaned dataset was divided into 35,179 training records and 8,795 testing records using an 80:20 stratified split. The TF-IDF vectorizer was fitted only on training data to avoid data leakage.

### 15. Model Validation

A hold-out test set was used to evaluate generalization. The selected Linear SVM was additionally validated through five-fold cross-validation during GridSearchCV tuning. The best cross-validation F1-score was 99.64%, indicating consistently strong performance across validation folds.

### 16. Hyperparameter Tuning

GridSearchCV with five-fold cross-validation was used to tune the Linear SVM. F1-score was used as the optimization criterion. The best parameter was C = 10. The tuned model was

subsequently evaluated on the held-out test set and selected as the final model.

### 17. Model Evaluation

Accuracy, precision, recall, and F1-score were used to evaluate the classifiers. Accuracy measures overall correctness, precision measures the reliability of positive predictions, recall measures the ability to identify actual class members, and F1-score balances precision and recall. A confusion matrix and classification report were also generated for detailed evaluation.

### 18. Model Comparison

| Model | Accuracy | Precision | Recall | F1-Score |
|---|---|---|---|---|
| Logistic Regression | 99.27% | 99.08% | 99.41 % | 99.25 % |
| Multinomial Naive Bayes | 95.65% | 95.52% | 95.45 % | 95.48 % |
| Linear SVM | 99.66% | 99.62% | 99.67 % | 99.65 % |
| Tuned Linear SVM | 99.72% | 99.67% | 99.74 % | 99.71 % |

### 19. Final Model Selection

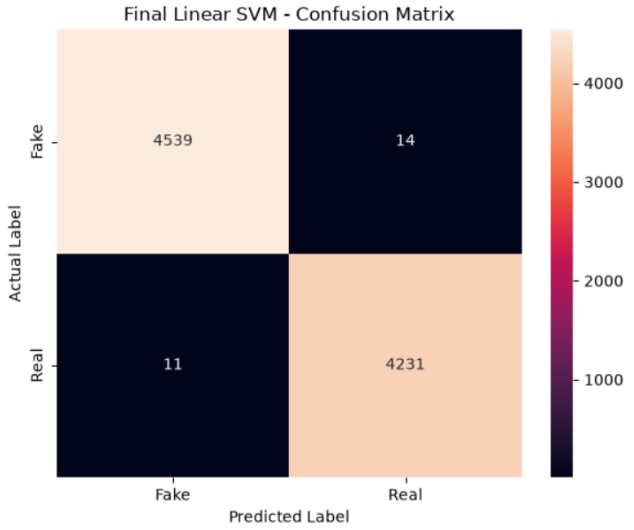
The Tuned Linear Support Vector Machine was selected because it achieved the strongest overall test performance and strong cross-validation results. Final performance was: Accuracy 99.72%, Precision 99.67%, Recall 99.74%, F1-score 99.71%, and best cross-validation F1-score 99.64%.

### 20. Sample / Unseen Data Prediction

The final model and TF-IDF vectorizer were saved using Joblib. For a new news article, the same cleaning procedure is applied, the text is transformed by the saved vectorizer, and the resulting feature vector is classified by the saved SVM. The label mapping is 0 for Fake News and 1 for Real News. The notebook also demonstrated prediction on unseen examples.

### 21. Results

The project successfully completed the full workflow from raw data loading to model persistence and unseen-news prediction. TF-IDF produced an effective representation for the news text. Logistic Regression achieved very high performance, Multinomial Naive Bayes provided a useful baseline, and Linear SVM achieved the strongest results. Tuning further improved the Linear SVM, which achieved 99.72% test accuracy and 99.71% F1-score.




### 22. Discussion

The results show that traditional machine learning can perform extremely well on this dataset when combined with appropriate text preprocessing and TF-IDF features. Linear SVM outperformed the other evaluated models, likely because of its ability to create an effective decision boundary in a high-dimensional sparse feature space. Results should nevertheless be interpreted carefully because performance may vary for news from different sources, topics, and time periods.

### 23. Limitations

The dataset represents a particular collection of articles and may not include all modern misinformation patterns. The model depends on textual patterns learned from training data and does not independently verify factual correctness. TF-IDF does not capture deep semantic context like transformer-based models. The project also does not incorporate images, source credibility, or social-network information.

## 24. Conclusion

This project developed a complete machine learning system for Fake and Real News Detection. After combining and cleaning the datasets, TF-IDF with up to 50,000 unigram and bigram features was used for text representation. Logistic Regression, Multinomial Naive Bayes, and Linear SVM were compared. The Linear SVM was tuned using GridSearchCV and five-fold cross-validation, with C = 10 selected as the best parameter. The tuned Linear SVM achieved 99.72% accuracy, 99.67% precision, 99.74% recall, and 99.71% F1-score. The trained model and vectorizer were saved and successfully reused for unseen-news prediction.

## 25. Future Scope

- Evaluate character n-grams and word n-grams to capture additional message patterns.
- Preserve selected numerical and symbolic features that may be useful for spam detection.
- Use automated hyperparameter tuning when baseline results indicate meaningful room for improvement.
- Compare the current approach with deep learning and transformer-based language models.
- Develop a simple web interface or API for real-time SMS classification.
- Periodically retrain the model using newer messages to adapt to changing spam strategies.
- Add confidence scores and human review for uncertain predictions.

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## 27. Appendix

### A. Dataset Summary

| Dataset | Records |
|---|---|
| Fake News Dataset | 23,481 |
| Real News Dataset | 21,417 |
| Original Combined Dataset | 44,898 |
| Cleaned Dataset | 43,974 |

**B. Train-Test Split**

| Split | Records |
|---|---|
| Training | 35,179 |
| Testing | 8,795 |

**C. Important Configuration**

- TF-IDF maximum features = 50,000
- N-gram range = (1,2)
- Cross-validation = 5 folds
- Best Linear SVM parameter = C = 10
- Final model = Tuned Linear SVM

**D. Final Results**

| Metric | Score |
|---|---|
| Accuracy | 99.72% |
| Precision | 99.67% |
| Recall | 99.74% |
| F1-Score | 99.71% |
| Best CV F1 | 99.64% |